

Education

- 2022-Present **Ph.D. of Computer Science**, *University of California-Davis*, United States
○ Advisor: Dr. Hamed Pirsiavash
- 2021-2022 **Ph.D. of Computer Science**, *University of Maryland-Baltimore County*, United States
TRANSFERRED
- 2015-2020 **Bachelors in Computer Engineering**, *University of Tehran*, Iran

Experience

- sept 2025-present **Netflix**, *Research Intern*, Los Gatos, CA
- Sept 2022-Sept 2025 **UC-Davis**, *Research student*, Davis, CA
- May 2022-Sept 2022 **Apple Inc.**, *Research Intern*, Seattle, WA
- May 2022-Aug 2021 **UMBC**, *Research Student*, Baltimore, MD
- Aug 2018-Aug 2019 **IPM: Institute For Research In Fundamental Sciences**, *Research Intern*, Tehran, Iran
- Aug 2017-Aug 2018 **University of Tehran**, *Research Student*, Tehran, Iran

Publications

FAT-RABBIT: Fault-Aware Training towards Robustness Against Bit-flip Based Attacks in Deep Neural Networks, *ITC 2024*

Developed a Deep Neural Network training method to overcome bit-flip based attacks

MCNC: Manifold-constrained reparameterization for neural compression, *ICLR 2024*

A novel approach designed to reparameterize PEFT for further compression and accuracy

Brainwash: A poisoning attack to forget in continual learning, *CVPR 2024*

An attack designed for inducing forgetting for continual learning methods

NOLA: Networks as linear combination of low rank random basis, *ICLR 2024*

A simple upgrade on top of LoRA fine-tuning for further compression

PRANC: Pseudo RAndom Networks for Compacting deep models, *ICCV 2023*

Introduced a re-parameterization of neural network that enables +100x reduction in memory

Sparsity and heterogeneous dropout for continual learning in the null space of neural activations, *CoLLA 2022*

Designed a method to increase the model capacity for continual learning approaches

A simple baseline for low-budget active learning, *Arxiv, 2021*

Introduced a baseline for active-learning in extremely low budget

Taxonn: A light-weight accelerator for deep neural network training, *ISCAS, 2020*

Designed and implemented a hardware architecture that enables training over the baseline implementation with minimal power and area overhead

Patents

2024 Taxonn: a light-weight accelerator for training deep neural networks on the edge

Skills

Programming Languages C, C++, Java, Python, Javascript, HTML, CSS, GoLang

Machine Learning PyTorch, Tensorflow, Numpy, Pandas

Web App Development React Native, ReactJS, Django, NodeJS, SQL

Hardware Design Verilog, Quartus, ModelSIM

Edge Device ESP modules, Arduino, AVR Microcontrollers, Raspberry PI, Cyclone FPGAs

Honors and Awards

2014 Bronze Medal in Iranian National Olympiad in Informatics

2013 Finalist in Khwarazmi festival

Notable Projects

PRANC Map-Reducer, Designed a Map Reducer for huge matrix multiplication for PRANC

Topic Labeling, Designed a training-free topic labeling using the university archive data

Obstacle Avoidance Robot, Designed a system for RC cars to avoid collision with obstacles

KhaneBDoosh, Student Project, Designed a web application with Android, iOS, and web front end for browsing and scheduling appointment for open house

Personal Surveillance Camera, Designed a whole pipeline of camera, server, and application for household security